

CODE CORTEX

AI-First Software Studio

Intelligent Products. Enterprise Platforms. Real AI.

35+	6	AI
Projects	Industries	Built into every product

WHAT WE BUILD

Solutions for Your Industry

Ready-to-adapt products, proven in real workflows, built to be customised to your business.

Education

Run a whole school from one system: admissions, fees, grades, timetables, parent portal and AI early warnings.

CC-EMS · SafeKid

Enterprise & Services

HR, projects, CRM, finance and approvals for a growing company, with audit trail built in.

CC-IMS

Retail

A point of sale that keeps selling offline, tracks stock and customer credit, and reports to the tax authority.

CC-POS

Smart City & Utilities

Citizen photo reports turned into prioritised jobs and optimised crew routes, with proof of cleanup.

CleanRoute-AI

Manufacturing

Catch machine problems early from vibration and temperature data, and inspect surfaces with a phone camera.

Anomaly Detection · Defect Detection

Transport & Mobility

Live van tracking, QR boarding checks and instant alerts so parents know their child is safe.

SafeKid

WHY US

Why Companies Choose Code Cortex

We Build Our Own Products

- We run real platforms, so we know what it takes to ship and support software
- Your project starts from proven building blocks, not a blank page

AI With Human Control

- AI drafts, scores and flags; people review before anything takes effect
- Every AI action can be logged, limited and switched off

Enterprise-Grade Foundations

- Role-based access, audit logs and two-factor sign-in
- Multi-tenant data isolation, backups and deployment guides
- Documented APIs for your other systems

Direct Access to Engineers

- Founder and engineers on every project
- Weekly check-ins, clear milestones
- No layers of account managers

Start Small, Grow With Us

- A small paid milestone to prove fit first
- Project, dedicated-team or long-term models
- Ongoing maintenance and support

Built for Emerging and Global Markets

- Urdu and Arabic right-to-left support, 23 languages in our ERP
- Local tax, payment and workflow needs
- Cost-efficient delivery from Pakistan

AI

AI Capabilities

Practical machine learning that solves business problems.

Computer Vision

- Detect objects and defects in photos and live video
- Fill-level and waste-type estimation
- Phone-camera inspection
- OpenCV, YOLO, MobileNetV2

Prediction & Anomaly Detection

- Risk scoring and early warnings
- Machine health from vibration and temperature
- Demand and complaint forecasting
- scikit-learn, XGBoost, Isolation Forest

Optimisation & Automation

- Route planning on real road networks
- Rule-based timetable generation
- Smart dispatch and alerts
- Google OR-Tools, OSRM

Language AI

- Drafted reports, messages and summaries
- On-demand translation
- Assistants that answer from your own data
- Works with leading LLM providers

Learning From Use

- Staff corrections become training data
- Scheduled retraining
- Measured accuracy reported honestly
- Human approval for consequential actions

Delivery Stack

- Python, Django, FastAPI, NestJS
- React, Next.js, Flutter
- PostgreSQL, Firebase, Docker
- Cloud deployment and CI/CD

EDUCATION

CC-EMS: School Management Platform

Everything a school needs in one system, with a portal for every role and AI that helps staff without replacing them.

Multi-tenant
one platform, many schools

23
languages incl. Urdu and Arabic

3
portals: parent, teacher, student

AI
drafts reviewed by staff

Academics & Admissions

- Online admissions and enrollment
- Gradebook, GPA, report cards, transcripts
- Timetable generator with rules and undo
- Exams, marksheets, homework, behaviour

Finance & Campus

- Fee plans, invoices, online payments
- Defaulter list and automatic reminders
- Payroll and salary slips
- Library, transport, inventory, cafeteria

Communication & Insight

- Attendance with instant parent alerts
- Announcements, messaging, meetings
- Dropout and fee-default early warnings
- Dashboards and CSV / Excel import-export

Security & Scale

- Role-based access and audit logs
- Authenticator-app 2FA
- Complete school data isolation
- Subscription plans and platform-owner console

ENTERPRISE

CC-IMS: Business Management System

HR, projects, sales and finance on one platform, with approvals and an audit trail for every important action.

1
system for HR, CRM, finance

Approvals
maker-checker and delegation

Audit
trail on every key action

Your cloud
or in-house server

People & HR

- Employee records, attendance, leave
- Payroll, payslips, loans
- Recruitment and onboarding
- Appraisals, goals and OKRs

Projects & Sales

- Projects, Kanban, sprints, Gantt
- Timesheets and capacity planning
- Leads, deals, quotations, contracts
- Client portal

Finance & Control

- Invoices, recurring and milestone billing
- General ledger and financial statements
- Bank reconciliation, vendor bills
- Anomaly flags and AI receipt reading

RETAIL

CC-POS: Point of Sale for Shops

Built for Pakistani retailers: it keeps selling when the internet or power goes out, and it tracks stock and customer credit.

Offline
first, syncs when back online

EN + UR
English and Urdu

FBR
invoice QR and reporting

Phone
becomes the barcode scanner

Sell Faster

- Barcode, weight-based and quick-pick selling
- Cash, card, wallet and khata (credit)
- Returns, layaway, quotes, gift cards
- WhatsApp and thermal-printer receipts

Control Stock & Cash

- Low-stock and expiry alerts
- Receiving, stocktake, transfers, purchasing
- Supplier payables, cash office, Z report
- Staff roles and tamper-evident audit trail

Grow With Insight

- Forecast, price watch, shrinkage
- Expiry rescue and khata-risk alerts
- Online ordering page for customers
- Multiple branches with cloud sync

SMART CITY

CleanRoute-AI: Smart Waste Management

Citizens report waste with a photo. AI judges it, ranks it, sends the nearest crew, and checks the job was done.

Photo
to priority in seconds

Road-based
route optimisation

Verified
cleanup with photo + GPS

Learns
from every admin correction

Report & Prioritise

- Photo report with location
- Fill level and waste-type detection
- Priority raised near schools, hospitals, mosques, repeat reports, rain
- Blur, duplicate-photo and wrong-location checks

Dispatch & Optimise

- Smart dispatch into the nearest crew route
- Google OR-Tools with real road distances
- Crew live location and daily capacity
- Automatic escalation of overdue jobs

Verify & Improve

- After-photo and GPS to close a job
- Auto-approval or admin review queue
- Forecasts, hotspots and spike alerts
- SLA, crew and AI reports with CSV export

TRANSPORT

SafeKid: Student Transport Safety

Parents see where their child is, from pickup to school and back home, with an instant alert at every step.

Live GPS
van location and ETA

QR
scan on boarding and drop-off

Instant
alerts to parents

3 roles
parent, driver, assistant

For Drivers & Schools

- Van and student setup with pickup stops
- Unique QR Safety Pass per student
- Morning and afternoon trips
- Parent ratings feed a performance dashboard

For Parents

- Clear status: at home, in transit, arrived
- 'Ready for Pickup' button
- Live van map and arrival time
- Fee invoices and payment history

Safety & Communication

- Boarding and drop-off logged with time and place
- Driver-parent messaging
- AI assistant for common questions
- Google, phone OTP or biometric sign-in

MANUFACTURING & INDUSTRY

Industrial AI: Detect Problems Early

Machine-health monitoring and visual inspection, ready to pilot on your data.

Early
warning before failure

Phone
camera inspection

API
+ dashboard included

Pilot
on your own data

Machine Anomaly Detection

- Learns normal vibration and temperature behaviour
- Flags unusual patterns with an anomaly score
- Measures precision, recall, false alarms and lead time before failure
- Cleans messy sensor files; maps your column names
- REST API and live dashboard

Visual Defect Inspection

- Detects six types of steel surface defects
- Works from a phone camera with live scan
- Handles large photos by tiling
- Adapts lab-trained models to phone images
- Annotated results with confidence scores

Validated on public benchmark data; the next step is a short pilot on your own plant data to prove value before any larger commitment.

PARTNER

How We Work With You

A low-risk path from idea to production.

1

Understand

Short review of your system, data and goals.

2

Pilot

One agreed piece of work with a fixed scope, price and written acceptance tests.

3

Build

Weekly check-ins, clear milestones, modular architecture.

4

Support

Deployment, documentation, handover and ongoing maintenance.

Project-Based

- Defined scope and deliverables
- Fixed or milestone pricing

Dedicated Team

- Extra engineering capacity for your team
- White-label and overflow builds

Long-Term Partner

- Product evolution and maintenance
- Priority support

Confidentiality, code ownership and IP terms are agreed in writing before work starts.



Let's Build Something Intelligent

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